

F-16 BVR COMBAT RL AGENT

Beyond Visual Range Air Combat with Reinforcement Learning
Technical Architecture & Implementation Report

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STATUS	DETAIL
WVR Agent	86% kill rate vs LAG_PURSUIT (validated)
BVR Simulation	15/15 tests passing at 43× realtime on Windows Python 3.9
BVR Training	Running – diagnosing 0% win rate at episode ~3000
GUI	Fully operational – Three.js 3D tactical view, offline-capable

1. PROJECT OVERVIEW

This project develops an RL agent for **Beyond Visual Range (BVR)** 1v1 F-16 air combat. It extends a WVR system (86% kill rate) to the harder BVR domain where engagements span 20–80 km with radar-guided missiles requiring continuous datalink.

Key Decisions

- Pure-Python simulation – no C++, no DLL, no ports. 43× realtime throughput.
- MaskablePPO – action masking blocks physically impossible missile launches
- $\gamma \geq 0.997$ – at $\gamma=0.95$ only 13% of kill reward reaches the launch decision
- Asymmetric actor-critic – actor sees radar only (partial obs), critic sees full truth
- 6-stage curriculum: STRAIGHT → EVASIVE → NOTCHER → SHOOTER → ADAPTIVE → SELF-PLAY

Aspect	WVR	BVR
Range	< 5 km	20–80 km
Weapon	Gun/IR missile	AIM-120 AMRAAM
Decision rate	10 Hz	1 Hz
Episode length	30–60 s	200–300 s
Gamma required	0.99	≥ 0.997
Obs dims	18	46
Status	86% kill rate ✓	Training in progress

2. SIMULATION ARCHITECTURE

Pure-Python hierarchy: world integrator calls F-16 and missile sub-models, returns a telemetry dict identical to original C++ UDP packets. No external dependencies.

2.1 F-16C Flight Model (f16_sim.py)

- Point-mass 3-DOF with analytical attitude propagation and body rate transport correction
- SEP ~47 m/s at 9000 m M0.9 · Turn rate ~5 °/s sustained (real: 8-12 °/s)
- Autopilot: proportional heading→bank→nz, altTarget→FPA, V→throttle

2.2 AIM-120 Missile (missile_sim.py)

- Proportional Navigation guidance N=4; guides on RADAR ESTIMATE, not truth
- 3-second datalink support rule; seeker handoff at random U(8,16) km
- Miss detection: if relative velocity sign flips before impact, miss is logged

2.3 World Integrator (sim_world.py)

```
obs = world.step(cmd_blue, cmd_red) # 1 call = 1 sim-second
```

Metric	Value
Throughput	43× realtime (39-43× measured, Windows Python 3.9)
Decision rate	1 Hz (1 step = 1 policy decision = 1 sim-second)
Parallelism	n_envs × 43× (SubprocVecEnv, one process per env)
Episode length	300 decisions = 300 sim-seconds
Tests	15/15 pass (test_sim.py)

3. RL ENVIRONMENT (bvr_env.py)

Action Space – MultiDiscrete(10×5×4×2)

Sub-action	Bins	Values
Heading offset	10	−90° to +90° in ±10° steps (0°=hot, ±50°=crank, ±90°=beam, 180°=dra
Altitude delta	5	−3000 m to +3000 m
Speed target	4	220 / 260 / 300 / 340 m/s
Fire missile	2	0=hold / 1=launch (masked unless in envelope + TRACK)

Observation Space – 46 Dimensions

Group	Dims	Contents
Ownship kinematics	8	pos/vel ENU, heading, bank, nz, speed, alt, fuel
Track state	6	State enum, σ_{pos} , σ_{vel} , age, target pos+vel estimate
Envelope scalars	4	r/Rmax (own), r/NEZ (own), r/Rmax (threat), r/NEZ (threat)
Missile state	8	TGO_est, needs_support, t_stale, seeker_active (per missile)
RWR	3	launch_warn, seeker_active_inbound, bearing
Geometry	17	Range, closure, HCA, aspect angles, F-pole, fuel

Reward Structure

Event	Reward
KILL	+1.0
KILLED	−1.0
MUTUAL KILL	−0.4
TIMEOUT (passive collapse)	−0.9
Potential shaping	±0.05/step (envelope, track quality, energy, support)

4. TRAINING PIPELINE

MaskablePPO (sb3-contrib)

- $\text{Gamma} = 0.997 \cdot \text{LR} = 2.5\text{e-}4 \cdot \text{batch} = 256 \cdot \text{n_steps} = 1024$
- Asymmetric AC: actor sees 46D radar obs; critic sees 16D privileged truth

Curriculum Stages

#	Stage	Key Challenge	Advance Criterion
1	STRAIGHT	Search, approach, track, fire	Win rate $\geq 65\%$ for 50 ep
2	EVASIVE	Maintain radar lock during manoeuvres	Win rate $\geq 65\%$ for 50 ep
3	NOTCHER	Counter Doppler beam notch tactics	Win rate $\geq 65\%$ for 50 ep
4	SHOOTER	Counter full BVR doctrine	Win rate $\geq 65\%$ for 50 ep
5	ADAPTIVE	Generalise vs randomised tactics	Win rate $\geq 65\%$ for 50 ep
6	SELF_PLAY	Agent vs own archived checkpoints	

Time Estimates (43× realtime / env)

Steps	n_envs=1	n_envs=2	n_envs=4	n_envs=8
500k	~3 h	~1.5 h	~45 min	~22 min
2M	~13 h	~6.5 h	~3.2 h	~1.6 h
5M	~32 h	~16 h	~8 h	~4 h

5. GUI (bvr_gui.py + bvr_gui.html)

655 KB self-contained HTML. Three.js r128 embedded inline. Works fully offline. Replicates WVR viz visual style: same sky shader, ground color, fog, lighting, jet shape.

```
python bvr_gui.py # HTTP :5006 WebSocket :5010
```

3D View – Three.js WebGL

- Sky shader gradient: 0x2f74c0 (top) → 0xcfe2ef (horizon)
- Ground: 0x6e5a3c (brown), FogExp2(0xbcd6e8, 0.0000085)
- HemisphereLight(0xbfe0ff, 0x6b5838, 1.0) + DirectionalLight(0xffff6e6, 1.15)
- F-16 planform: nose=[1.6,0,0], wings=[-0.6,±0.9,0], S=150 scale + white edge outlines
- Vertex-coloured trails (25→100% brightness fade, same as WVR viz)
- Missiles with TGO arc: green→amber→red as support expires; teal ring on seeker-active
- Camera: drag=orbit, scroll=zoom (0.08–20×), double-click=follow aircraft

Key Metrics & Diagnosis

Metric	Target	Warning Signal
Win rate	> 65% per stage	< 30% after 5k ep → check reward design
Timeout rate	< 20%	> 30% → passive collapse, raise R_TIMEOUT
Realized Pk	0.6 – 0.9	< 0.3 → shots outside envelope
r/Rmax at shot	0.5 – 0.9	> 1.0 → Rmax overestimated
Track fraction	> 70%	< 40% → excessive cranking past gimbal limit
Exchange ratio	> 2.0 vs SHOOTER	< 1.0 → more losses than kills

6. CURRENT STATUS & NEXT STEPS

Completed

Item	Status
Pure-Python simulation	15/15 tests pass, 43× realtime, Windows Python 3.9.7
RL environment	Action masking, asymmetric critic, potential shaping, 46D obs
Training pipeline	MaskablePPO + 6-stage curriculum + auto-advance at 65%
GUI	Three.js 3D view, full metrics, controls, shot trace table
WVR baseline	86% kill rate vs LAG_PURSUIT – architecture validated

Active Issue: 0% Win Rate at Episode ~3000

Observed: timeout rate 46% (high), $r/R_{max}=1.071$ at shot, 0% win vs STRAIGHT (unarmed).

Diagnosis: Run eval episode. If trails diverge → passive collapse (raise `R_TIMEOUT` to `-0.9`). If missile misses → R_{max} overestimated (lower `R_MAX_REF` to 50000 in `bvr_envelope.py`).

Next Steps

Priority	Task
HIGH	Run eval → diagnose passive collapse vs kinematic miss
HIGH	Apply fix (<code>R_TIMEOUT</code> or <code>R_MAX_REF</code>) → resume from <code>models_bvr/latest</code>
MED	Run <code>sweep_envelope.py</code> to calibrate R_{max}/NEZ vs real missile kinematics
MED	Scale to <code>n_envs=4</code> once STRAIGHT curriculum is passing
LOW	Checkpoint A/B comparison; opponent heatmap visualisation

File Reference

File	Purpose
<code>f16_sim.py</code>	F-16C point-mass model
<code>missile_sim.py</code>	AIM-120 PN guidance + datalink
<code>sim_world.py</code>	World integrator
<code>bvr_env.py</code>	Gymnasium RL environment
<code>bvr_track_adapter.py</code>	EKF radar track adapter
<code>bvr_radar_sim.py</code>	Python radar surrogate (53 μ s/call)
<code>bvr_envelope.py</code>	AIM-120 launch envelope model
<code>bvr_opponents.py</code>	Scripted opponents (5 types)
<code>bvr_policy.py</code>	Asymmetric actor-critic policy
<code>train_bvr.py</code>	MaskablePPO training script

bvr_gui.py	HTTP+WebSocket GUI server
bvr_gui.html	Three.js browser frontend (655 KB, offline)
test_sim.py	Integration test suite (15 tests)